

BIOS15: LAB MANUAL



Harmony Folse
Chabot College

BIOS15 Lab Manual

This text is disseminated via the Open Education Resource (OER) LibreTexts Project (<https://LibreTexts.org>) and like the thousands of other texts available within this powerful platform, it is freely available for reading, printing, and "consuming."

The LibreTexts mission is to bring together students, faculty, and scholars in a collaborative effort to provide an accessible, and comprehensive platform that empowers our community to develop, curate, adapt, and adopt openly licensed resources and technologies; through these efforts we can reduce the financial burden born from traditional educational resource costs, ensuring education is more accessible for students and communities worldwide.

Most, but not all, pages in the library have licenses that may allow individuals to make changes, save, and print this book. Carefully consult the applicable license(s) before pursuing such effects. Instructors can adopt existing LibreTexts texts or Remix them to quickly build course-specific resources to meet the needs of their students. Unlike traditional textbooks, LibreTexts' web based origins allow powerful integration of advanced features and new technologies to support learning.



LibreTexts is the adaptable, user-friendly non-profit open education resource platform that educators trust for creating, customizing, and sharing accessible, interactive textbooks, adaptive homework, and ancillary materials. We collaborate with individuals and organizations to champion open education initiatives, support institutional publishing programs, drive curriculum development projects, and more.

The LibreTexts libraries are Powered by [NICE CXone Expert](#) and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation nor the US Department of Education.

Have questions or comments? For information about adoptions or adaptations contact info@LibreTexts.org or visit our main website at <https://LibreTexts.org>.

The LibreTexts name, logo, and book covers are trademarked by LibreTexts, Inc. (a 501(c)(3) not-for-profit organization) and protected against unauthorized use.

This text was compiled on 08/07/2026

- [InfoPage](#) by [Delmar Larsen](#) is licensed [Public Domain](#).

TABLE OF CONTENTS

Acknowledgements

About the Author

Licensing

1: Language of Anatomy

- 1.1: Introduction to Anatomical Terms
- 1.2: Surface and Regional Anatomy
- 1.3: Body Orientation and Direction
- 1.4: Planes and Sections
- 1.5: Body Cavities
- 1.6: Reviewing Anatomical Language

2: Membrane Transportation

- 2.1: Introduction to Membrane Transportation
- 2.2: Diffusions of Crystals through an Agar Gel
- 2.3: Osmosis
- 2.4: Practice Graphing
 - 2.4.1: Practice Graph Results
- 2.5: Procedure for Osmosis Experiment
- 2.6: Results for Diffusion and Osmosis Experiments
 - 2.6.1: Graph for your Osmosis Experiment

3: Microscopy

- 3.1: Introduction to Microscopy
- 3.2: Microscope Components
- 3.3: Microscope Handling and Care
- 3.4: Total Magnification
- 3.5: Field of View
- 3.6: Practicing Microscopy
- 3.7: Practice Labelling the Microscope

4: Epithelial and Connective Tissues

- 4.1: Introduction to Tissues
- 4.2: Epithelial Tissues
- 4.3: Connective Tissues
- 4.4: Microscope Slides - Epithelial and Connective Tissue
- 4.5: Questions and Review - Epithelial and Connective Tissues

5: The Axial and Appendicular Skeleton with Joint Movement

- 5.1: Introduction to the Skeletal System
- 5.2: Microscope Slides - Compact Bone
- 5.3: The Skull
- 5.4: The Vertebral Column

- 5.5: The Thorax
- 5.6: Bones of the Pectoral Girdle and Upper Limbs
- 5.7: Bones of the Pelvic Girdle and Lower Limbs
- 5.8: Joint Movement
- 5.9: Questions and Review - Skeletal System and Joints

6: Muscle Tissue

- 6.1: Introduction to Muscle Tissue
- 6.2: Structures of Muscle Tissue
- 6.3: Microscope Slides - Muscle Tissue
- 6.4: Questions and Review - Muscle Tissue

7: Axial and Appendicular Skeletal Muscles

- 7.1: Introduction to Skeletal Muscle
- 7.2: Muscles of the Head and Neck
- 7.3: Muscles of the Axial Skeleton
- 7.4: Muscles of the Upper Limb
- 7.5: Muscles of the Hip and Lower Limb
- 7.6: Questions and Review - Skeletal Muscles

8: Nervous System

- 8.1: Introduction to the Nervous System
- 8.2: Neuron Anatomy
- 8.3: Brain Anatomy - Cerebrum and Cerebellum
- 8.4: Brain Anatomy - Diencephalon and Brain Stem
- 8.5: Cranial Nerves
- 8.6: Spinal Cord Anatomy
- 8.7: Microscope Slides - Brain and Spinal Cord
- 8.8: Sheep Brain Dissection
- 8.9: Questions and Review - Nervous System

9: Reflexes

- 9.1: Introduction to Reflexes
- 9.2: Reflex Tests
- 9.3: Questions and Review - Reflexes

10: Special Senses - Eyes and Ears

- 10.1: Introduction to Eyes and Ears
- 10.2: Eye Anatomy
- 10.3: Cow Eye Dissection
- 10.4: Ear Anatomy
- 10.5: Microscope Slides - Eyes and Ears
- 10.6: Questions and Review - Eyes and Ears

11: General and Special Sensory Tests

- 11.1: Introduction to General and Special Sensory Tests
- 11.2: Sensory Tests on the Eye
- 11.3: Sensory Tests on the Ears
- 11.4: General Sensory Tests

- [11.5: Results of Sensory Tests](#)

12: Endocrine System

- [12.1: Introduction to the Endocrine System](#)
- [12.2: Microscope Slides - Endocrine Glands](#)
- [12.3: Questions and Review](#)

13: Blood

- [13.1: Introduction to Blood](#)
- [13.2: Types of Blood Cells](#)
- [13.3: Microscope Slides - Blood Cells](#)
- [13.4: Hematocrit](#)
- [13.5: Blood Typing](#)
- [13.6: Results and Questions](#)

14: Heart Anatomy

- [14.1: Introduction to the Heart](#)
- [14.2: Heart Anatomy](#)
- [14.3: Sheep Heart Dissection](#)
- [14.4: Questions and Review](#)

15: Electrocardiograms (ECGs/EKGs)

- [15.1: Introduction to Heart Electrophysiology](#)
- [15.2: Instructions for Recording an ECG](#)
- [15.3: ECG Results and Review](#)

16: Blood Pressure

- [16.1: Introduction to Blood Pressure Measurements](#)
- [16.2: Instructions for Measuring Blood Pressure](#)
- [16.3: Results and Questions](#)

17: Respiration and Spirometry

- [17.1: Introduction to Respiration](#)
- [17.2: Respiratory System Anatomy](#)
- [17.3: Microscope Slides - Respiratory System](#)
- [17.4: Spirometry and Measuring Lung Capacity](#)
- [17.5: Questions and Review](#)

[Index](#)

[Glossary](#)

[Detailed Licensing](#)

[Detailed Licensing](#)

Acknowledgements

This OER was developed with the support of the [ZTC Grant Program](#) funded by the California Community Colleges Chancellor's Office. It is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License](#) (CC BY-NC 4.0), which means it's free to use, adapt, and share for educational purposes.

The author would also like to thank the following:

- Dr. Cristina Moon at Chabot College, for her guidance and for sharing her expertise and knowledge on creating OER resources.
- Megan Jensen at Chabot College, for her guidance in starting this project.